Malaysian Genomics Resource Centre

Certificate of Analysis

Report for Exosomes

Batch Number	
Manufacturing Date	
Expiry Date	
Dosage Form	Clear-colourless solution
Storage Temperature	4 °C

TEST	RESULTS
Sterility Test	QUALITY ASSURANCE PASSED
Limulus Amoebocyte Lysate Test	QUALITY ASSURANCE PASSED
Mycoplasma	QUALITY ASSURANCE PASSED
Immunophenotyping	QUALITY ASSURANCE PASSED
Nanoparticle Tracking Analysis (NTA)	QUALITY ASSURANCE PASSED
Next Generation Sequencing (NGS)	QUALITY ASSURANCE PASSED

This certifies that the exosomes delivered by Malaysian Genomics Resource Centre Berhad is deemed suitable for human use as it has been tested, verified and approved in the tests above.

Storage recommendation:
1) Refrigerate the vials in 4°C at all times upon receival
2) Keep away the vials from direct heat or sunlight
3) Avoid prolong exposure in room condition
4) It is recommended to place the vials in -20°C for prolonged storage

Authorised Signature:

Malaysian Genomics Resource Centre

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	TEST	SPECIFICATIONS	TEST RESULTS	REMARKS
1.	Sterility	No growth (no turbidity) in the test media	Non-turbid	QC passed
2.	Limulus Amoebocyte Lysate Test	Endotoxin sensitivity at 0.125	Yes	QC passed
3.	Mycoplasma	Mean of absorbance <0.05	0.03	QC passed
4.	Immunophenotyping	Marker – CD63, CD9, CD81 (≥ 85%).	Within the range	QC passed
5.	Nanoparticle Tracking Analysis (NTA)	Exosome particle size range 30nm – 200nm	9×10^9 particles/mL	QC passed
6.	Next Generation Sequencing (NGS)	Exosomal miRNA	hsa-miR-205, hsa-miR-155, hsa-miR-99, hsa-miR-146, hsa-miR-218, hsa-miR-4530	QC passed

Authorised Signature:

Supporting Information

STERILITY TEST [5, 6]

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days. Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

MYCOPLASMA TEST [4,5]

Negative findings for mycoplasma species *M. hyorhinis, M. arginini, M. fermentans, M. orale, M. Pirum, M. hominis, M. salivarium and A. laidlawii.*

LIMULUS AMOEBOCYTE LYSATE TEST [3,5]

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

IMMUNOPHENOTYPING [1,2]

Antibodies	Results on Exosomes
CD9	+
CD63	+
CD81	+

* '+' indicates the presence of the marker in the cell.

NANOPARTICLE TRACKING ANALYSIS (NTA) [7,8]

Established method for the direct and real-time visualization and analysis of the nanoparticles size distribution and concentration in liquids.

NEXT GENERATION SEQUENCING (NGS)[9]

miRNA	Function	Regulation
hsa-miR-205	Migration and proliferation of epithelial cells[10]	Up-regulated
hsa-miR-155	Competitive fitness and proliferative potential of Foxp3, regulatory T cells[11]	Up-regulated
hsa-miR-99	Cell proliferation, cell migration[11]	
hsa-miR-146	Formation of new blood vessels[12]	Up-regulated
hsa-miR-218	Neovascularization[13]	Down-regulated
hsa-miR-4530	Angiogenesis[14]	Up-regulated

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